MILITARY SPECIFICATION

BEACON, RADAR AN/APN-154(V)

This specification has been approved by the Naval Air Systems Command, Department of the Navy

1. SCOPE

1.1 Scope - The equipment covered by this specification shall receive interrogations from X-band radars and transmit a reply to the interrogation on another frequency within the X-band.

*1.2 Classification - A Radar Beacon AN/APN-154(V) shall consist of the following items in the quantity indicated:

Item	Quantity
Receiver-Transmitter, Radar	1
Control	1
Duplexer	1
Cable Assembly, Radio Frequency	2

*1.3 Receiver-Transmitter, Radar - The Receiver-Transmitter shall be available in the following configurations. The particular configuration is to be specified in the contract or purchase order. (See paragraph 6.8)

Type Designation	Brief Description	Applicable Paragraph
RT-681/APN-154(V)	High band receiver - flange mount (see Figure 1)	3.4.1 Table I
RT-762/APN-154(V)	Low band receiver - side mount (see Figure 2)	3.4.1 Table I
RT-763/APN-154(V)	High band receiver - side mount (see Figure 2)	3.4.1 Table I
RT-764/APN-154(V)	Low band receiver - flange mount (see Figure 1)	3.4.1 Table I
RT-681A/APN-154(V)	High band receiver - flange mount and elapsed time indicator (see Figure 3)	3.4.1 Table I
RT-762A/APN-154(V)	Low band receiver - side mount with elapsed time indicator (see Figure 4)	3.4.1 Table I

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Type Designation	Brief Description	Applicable Paragraph
RT-763A/APN-154(V)	High band receiver - side mount with elapsed time indicator (see Figure 4)	3.4.1 Table I
RT-764A/APN-154(V)	Low band receiver - flange mount with elapsed time indicator (see Figure 3)	3.4.1 Table I

*1.4 Control - The Control shall be available in the following configurations, which shall be specified in the Contract or purchase order.

Type Designation	Brief Description	Applicable Paragraph
C-4419/APN-154(V)	Control less elapsed time indicator	3.4.2
C-4419A/APN-154(V)	Control with elapsed time indicator	3.4,5

*1.5 Duplexer - The Duplexer shall be available in the following configurations.

Type Designation	Brief Description	Applicable Paragraph
CU-1104/APN-154(V)	Duplexer - circulator 4-port	3.4.3

*1.6 Cable Assembly - The rf cable assemblies shall be available in the following configuration.

Type Designation	Brief Description	Applicable Paragraph
CG-2574/APN-154(V)	Rf cable - 6 inches long	3.4.4

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-W-5088	Wiring; Aircraft, Installation of
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-I-6181	Interference Control Requirements, Aircraft Equipment

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MIL-C-6781 Control Panel: Aircraft Equipment, Rack or Console Mounted

MIL-E-17555 Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of

MIL-T-18303 Test Procedures; Preproduction and Inspection, for Aircraft Electronic Equipment, Format for

MIL-N-18307 Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment

STANDARDS

Military

MIL-STD-704 Electric Power, Aircraft, Characteristics and Utilization of

2.2 Availability of Documents

(1) When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code DCS, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

3. REQUIREMENTS

*3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern:

(1) Parts and materials requirements shall conform to Specification MIL-E-5400.

(2) Nonrepairable subassemblies, as outlined in specification MIL-E-5400, shall be used when practicable. The general size of the subassembly, and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable. (See 6.7)

(3) Repairable subassembly design incorporated in previous procurements shall be changed to nonrepairable only upon approval of the procuring activity.

3.1.1 Non-Standard Parts and Materials Approved - Approval for the use of non-standard parts and materials shall be obtained as outlined in Specification MIL-E-5400.

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3.2 Design and Construction - The equipment shall conform with all the applicable requirements of Specification MIL-E-5400 for design, construction and workmanship, unless otherwise specified herein.

*3.2.1 Subminiature Construction - Semiconductor devices shall be used throughout the equipment to the greatest extent possible and the use of vacuum tubes shall be kept to a minimum. The modulator shall be of the semiconductor type. The use of the latest type silicon junction transistors shall be considered for this equipment.

*3.2.1.1 Subassemblies - The design, layout, and assembly of the subassemblies and their component parts shall be such as to facilitate quantity production and to result in minimum size and weight. Microminiature construction techniques shall be utilized to the fullest extent possible consistent with dependability, maintenance requirements, reliability, and the performance of this specification. The design shall include, when possible, circuits that have been or can be reproduced by mechanized or semi-mechanized production facilities consistent with the state of the art. The following types of construction are considered to be classed in the mechanized or semi-mechanized category.

(a) Subassemblies using printed circuits, upon which the parts are printed or mechanically placed and soldered.

(b) Construction in which integrated circuit techniques are utilized (process whereby circuit components such as transistors, diodes, resistors, capacitors, etc., are formed by use of semiconductor fabrication techniques, either separately or in combination) such that entire circuit stages are included in a single replaceable package. The contractor shall inform the procuring activity of those circuits he proposes to make as mechanized assemblies and shall obtain approval therefor.

*3.2.2 Total Weight - The total weight of the equipment shall be a minimum consistent with good design and will vary with the particular configuration specified in the contract. No configuration shall exceed 6.3 pounds.

3.2.3 Interference Control - The generation of radio frequency interference by the equipment and the vulnerability of the equipment to radio frequency interference shall be controlled within the limits of Specification MIL-I-6181, except that the rf output of any transmitter key up or receiver shall not exceed 60 db above 1 microvolt for cw or 70 db above 1 microvolt per megacycle for pulse cw interference at any frequency between 0.15 and 10,000 megacycles.

3.2.3.1 Receiver Front-End Rejection - Front end rejection of the receiver shall be equal to or greater than the limit shown in Figure 23 of Specification MIL-I-6181 except that at the image frequency, the rejection shall be at least 40 db.

3.2.4 Operating Life

3.2.4.1 Operational Stability - The equipment shall be capable of operating with optimum performance for a total of 150 hours continuously, or intermittently, without the necessity of readjustment of any controls which are inaccessible during flight.

3.2.4.2 Total Operating Life - The equipment shall have a minimum total operating life of 6,000 hours with reasonable servicing and replacement of parts. Parts requiring replacement within this interval, and the life of such parts, shall be specified by the manufacturer.

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3.2.4.3 Reliability in Mean Time Between Failures - The equipment shall have a minimum of 500 hours of mean (operating) time between failure when tested and accepted as outlined under the requirements of 4.4.3.

3.2.5 Control Panels - All rack or console mounted control panels shall conform to the applicable requirements of Specification MIL-C-6781. The configuration of all control panels must be approved by the procuring activity prior to preproduction testing.

3.2.6 Interchangeability - The equipment shall meet the interchangeability requirements as defined in Specification MIL-E-5400.

3.2.7 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in Specification MIL-E-5400. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

3.2.8 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307.

3.2.9 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient (25° $\pm 5^{\circ}$C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input Power Voltage	28 ±1 vdc

3.2.10 Primary Input Power Requirements - The equipment shall meet all applicable requirements of MIL-STD-704 and shall give specified performance from the following power sources with characteristics as defined in MIL-STD-704 having limits as specified therein. The power required shall not exceed the specified amounts.

(1) DC Power 28 V Category "B" 1.25 Amps

3.2.10.1 Under Voltage Protection - The equipment shall not be damaged by voltages below the minimum specified herein and shall automatically resume normal operation when the voltage returns within limits.

3.2.11 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions, or reasonable combination of these conditions, as specified in Specification MIL-E-5400 for Class 2 equipment, except as modified herein:

3.2.11.1 Altitude - The maximum altitude shall be increased to 100,000 feet.

3.2.11.2 Vibration - The equipment shall operate satisfactorily when subjected to the vibration requirements of Curve IV of Specification MIL-E-5400, with the frequency limit extended to 2000 cps. Vibration isolators shall not be used.

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3.2.11.3 Shock - When normally mounted, the transponder set shall be capable of withstanding, without damage, shock impact up to 100 g in any direction.

3.2.11.4 Acceleration - When normally mounted, the transponder set shall be capable of withstanding, without damage, constant acceleration of 100 g applied along any axis.

*3.2.11.5 Temperature - The equipment shall operate satisfactorily under the temperature vs. altitude conditions of Specification MIL-E-5400F for continuous operation except that Column II of Table II shall not apply. It should be noted that the temperatures specified herein are heat sink temperatures measured at the mounting flange of the receiver-transmitter.

3.2.12 Warm-Up Time - The time required for the equipment to warm up prior to operations shall be kept to a minimum and shall not exceed five minutes under standard conditions and fifteen minutes at extreme service conditions.

3.3 Performance

3.3.1 General - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

*3.3.2 Characteristics at X-Band - The equipment shall meet the performance standards as follows:

*3.3.2.1 Receiver - The center frequency of the receiver pass band shall be adjustable to any frequency from 8500 to 9600 mc by the installation of one of two interchangeable local oscillators. Each local oscillator shall be adjustable over a frequency range of not less than 500 mc.

*3.3.2.1.1 High Band Receiver - RT-681/APN-154(V), RT-681A/APN-154(V), RT-763/APN-154(V), and RT-763A/APN-154(V) configurations shall tune from 9000 to 9600 mc.

*3.3.2.1.2 Low Band Receiver - RT-762/APN-154(V), RT-762A/APN-154(V), RT-764/APN-154(V), and RT-764A/APN-154(V) configurations shall tune from 8500 to 9100 mc.

3.3.2.2 Transmitter - The transmitter frequency shall be adjustable to any frequency from 8800 to 9500 mc.

*3.3.2.3 Receiving and Transmitting Operation - The receiver-transmitter shall have separate receiver and transmitter antenna terminals such that it may be used with separate receiving and transmitting antennas. The transmitting antenna shall be separated from the receiving antenna such that at least 40 db of isolation shall occur.

3.3.2.4 Delay - Delay is defined as the time delay from the leading edge of the last pulse of the interrogating code group to the leading edge of the transmitted pulse.

3.3.2.4.1 Fixed Delay - The radar beacon shall have 1 delay setting. This shall be 1.0 ±0.1 usec. at a signal level of 0 dbm.

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3.3.2.4.2 Delay Variation with Signal Level - The total (absolute) variation in delay shall not exceed 0.2 usec. for signal levels of 0 dbm to within 3 db of threshold.

3.3.2.4.3 Delay Jitter - Delay jitter, which is defined as pulse to pulse variation in delay at a constant input signal level, shall not exceed the following values:

(a) For signal levels of 0 dbm to -55 dbm: 0.035 usec. peak-to-peak.

(b) For signal levels of -55 dbm to threshold: 0.05 usec. peak-to-peak.

3.3.2.5 Pulse Recurrence Frequency (PRF) - The equipment shall be capable of operation at any prf from 10 to 2,600 pps, unless this would result in exceeding the duty cycle limits specified in 3.4.1.7, in which case the maximum prf shall be that corresponding to the maximum allowable duty cycle.

3.3.2.6 Recovery Time - When operated in either the single pulse or coded interrogation mode, within the duty cycle limitations of 3.4.1.7, the radar beacon shall be capable of replying to each of 3 interrogations (code groups) when the spacing between any two successive interrogations is a minimum of 50 usec. These requirements shall be satisfied even though the signal level of adjacent interrogations differs by as much as 60 db. The peak power of the reply to each interrogation shall be within 1 db of the limits specified in 3.3.2.11. The radar beacon recovery time shall not be less than 13 usec.

3.3.2.7 Random Triggering - The equipment shall not generate random replies (those replies not synchronized with interrogations) at a rate greater than 10 pps under any operating conditions specified herein.

3.3.2.8 Sensitivity and Dynamic Range - The radar beacon shall reply to at least 99% of the interrogation signals at any input signal level between 0 dbm and -65 dbm under standard conditions. Under extreme conditions, sensitivity shall be at least -62 dbm. In the coded mode of operation, the sensitivity shall be measured at the optimum code spacing, which shall be within ±0.1 usec. of the nominal values defined in 3.4.1.3.2.

3.3.2.9 Overinterrogation Protection - The radar beacon shall not suffer permanent damage when subjected to interrogation of prf's which would result in exceeding the duty cycle limitation specified in 3.4.1.7. Within 5,000 usec., after the interrogation prf falls to the maximum specified value, or less, the radar beacon shall meet all requirements of this specification.

3.3.2.10 Impedance - The nominal input and output impedance of the radar beacon shall be 50 ohms.

3.3.2.11 Power Output - The peak power output of the radar beacon shall be 200 watts minimum under all conditions specified herein. Power output shall be measured at the antenna terminal of the duplexer-circulator.

3.3.2.12 Transmitter Frequency Stability - Under any operating conditions specified herein, except during changes in ambient temperature, the transmitter shall not drift more than ±5 mc and the drift rate shall not exceed 1 mc per minute after a 5 minute warm-up. During changes in ambient temperature, the temperature coefficient of frequency drift of the transmitter shall not exceed 50 kc per degree C. This shall include any frequency modulation present during service conditions as specified herein.

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3.4 Detail Requirements

*3.4.1 Receiver-Transmitter - There shall be eight configurations as listed in Table I. All configurations shall operate as described in the following paragraphs. The differences are in form factor, mounting, weight, tuning range, and inclusion of an elapsed time indicator. The contract shall specify the desired configuration.

Table I. Receiver-Transmitter Configurations

Item	Form Factor	Maximum Weight (lbs)	Tuning Range	Mounting	Elapsed Time Indicator
RT-681/APN-154(V)	Fig. 1	4.0	9000-9600 mc	Flange	No
RT-762/APN-154(V)	Fig. 2	4.4	8500-9100 mc	Side	No
RT-763/APN-154(V)	Fig. 2	4.4	9000-9600 mc	Side	No
RT-764/APN-154(V)	Fig. 1	4.0	8500-9100 mc	Flange	No
RT-681A/APN-154(V)	Fig. 3	4.3	9000-9600 mc	Flange	Yes
RT-762A/APN-154(V)	Fig. 4	4.7	8500-9100 mc	Side	Yes
RT-763A/APN-154(V)	Fig. 4	4.7	9000-9600 mc	Side	Yes
RT-764A/APN-154(V)	Fig. 3	4.3	8500-9100 mc	Flange	Yes

3.4.1.1 Receiver Frequency Stability - Under any operating conditions specified herein, except during changes in ambient temperature, the receiver frequency shall not vary more than ±3 mc after a 5 minute warm-up. During changes in ambient temperature, the coefficient of frequency drift shall not exceed 75 kc per degree C. However, changes in receiver frequency due to ambient temperature changes shall not require a change in the interrogating signal frequency in order to meet the sensitivity requirements of 3.3.2.8.

3.4.1.2 Selectivity

3.4.1.2.1 Bandwidth - The receiver bandwidth shall be not less than 8 mc nor more than the 14 mcs as measured at the points corresponding to a 3 db decrease from the maximum sensitivity.

3.4.1.2.2 Image - The image frequency shall be at least 100 mc from the nominal frequency. At the image frequencies the transponder shall exhibit 40 db decrease in sensitivity compared to that at the frequency to which the receiver is tuned.

3.4.1.3 Input Signal - The receiver-transmitter shall be capable of operation from several types of input signals depending on the code chosen. A choice of single pulse operation or operation with anyone of 5 double pulse codes shall be possible when used with the external (remote) control specified herein.

3.4.1.3.1 Single Pulse Operation - In this mode of operation the transmitter shall reply to a single pulse of the following characteristics:

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(a) Pulse Width — 0.3 to 3 usec.

(b) Rise Time — 0.1 usec. or less

(c) Frequency — Within ±2 mc of the frequency to which the receiver is tuned

3.4.1.3.2 Double Pulse Operation - The receiver-transmitter shall respond to any one of five spacings of double pulse interrogation as follows:

(a) 1.9 ±0.1 usec. - MODE 1

(b) 2.8 ±0.1 usec. - MODE 2

(c) 3.7 ±0.1 usec. - MODE 3

(d) 4.6 ±0.1 usec. - MODE 4

(e) 5.5 ±0.1 usec. - MODE 5

Individual pulse characteristics are:

(a) Pulse Width — 0.3 to 0.75 usec.

(b) Pulse Rise Time — 0.1 microsecond, or less

(c) Pulse Fall Time — 0.2 microsecond, or less

3.4.1.3.2.1 Double Pulse Rejection Characteristics

(1) When the radar beacon is set for acceptance of a double pulse interrogation whose spacing is 1.9 microseconds, interrogations having a pulse spacing less than 1.2 microseconds or greater than 2.6 microseconds shall not cause a beacon response to be generated.

(2) When the radar beacon is set for acceptance of a double pulse interrogation whose spacing is 2.8 microseconds, interrogations having a pulse spacing less than 2.1 microseconds or greater than 3.5 microseconds shall not cause a beacon response to be generated.

(3) When the radar beacon is set for acceptance of a double pulse interrogation whose spacing is 3.7 microseconds, interrogations having a pulse spacing less than 3.0 microseconds or greater than 4.4 microseconds shall not cause a beacon response to be generated.

(4) When the radar beacon is set for acceptance of a double pulse interrogation whose spacing is 4.6 microseconds, interrogations having a pulse spacing less than 3.9 microseconds or greater than 5.3 microseconds shall not cause a beacon response to be generated.

(5) When the radar beacon is set for acceptance of a double pulse interrogation whose spacing is 5.5 microseconds, interrogations having a pulse spacing less than 4.8 microseconds or greater than 6.2 microseconds shall not cause a beacon response to be generated.

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3.4.1.4 Input Impedance - The receiver terminal of the Receiver-Transmitter shall have a nominal input impedance of 50 ohms.

3.4.1.5 Transmitter Pulse Characteristics - The transmitter output pulse shall have the following characteristics:

(a) Pulse Width: 0.6 ±0.1 usec.

(b) Pulse Width Jitter: Not to exceed 0.01 usec. peak-to-peak at the one-half power points.

(c) Rise Time: Not to exceed 0.1 usec. from 10% to 90% amplitude points.

(d) Fall Time: Not to exceed 0.2 usec. from 90% to 10% amplitude points.

(e) Spectrum: The rf bandwidth (in megacycles) of the transmitter pulse, measured at one-quarter power by means of an rf spectrometer, shall not exceed 3.0 divided by the pulse width in microseconds.

3.4.1.6 Output Impedance - The transmitter terminal of the receiver-transmitter shall have a nominal output impedance of 50 ohms.

3.4.1.7 Transmitter Duty Cycle - The transmitter shall meet the requirements of this specification at any duty cycle up to .002.

3.4.1.8 VSWR of LOAD - The transmitter shall meet the requirements of this specification and shall not drift more than ±3/4 mc when operating into the Duplexer CU-1104/APN-154 with a load at the circulator antenna terminal having a mismatch such as to cause VSWR having a value varying up to 1.5 at any phase angle. In addition, the transmitter shall meet all of the requirements of this specification after application and removal of either a short circuit or an open circuit at the Duplexer-Circulator antenna terminal.

3.4.1.9 Function - The Receiver-Transmitter receives interrogations from X-band radars operating on the same frequency that the beacon receiver is tuned to; the interrogation signal is decoded and is used to trigger the beacon transmitter, which is tuned to another frequency within the X-band.

3.4.1.10 Contents - The Receiver-Transmitter shall include subassemblies that perform the following functions:

(1) Preselector

(2) Mixer

(3) Local Oscillator

(4) IF Amplifier

(5) Decoder

(6) Modulator

(7) Power Supply

(8) Magnetron

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*3.4.1.11 Form Factor - The form factor shall be as shown in Table I.

*3.4.1.12 Weight - The weight of this unit shall not exceed that shown in Table I.

3.4.1.13 Controls - The following controls shall be provided on the external surface of the Receiver-Transmitter.

(1) Preselector Frequency Control

(2) Local Oscillator Frequency Control

(3) Transmitter Frequency Control

Provision shall be made to prevent inadvertent movement of these controls. Adjustment of these controls shall require a hand tool of some type. Pressurized covers shall be provided to maintain receiver-transmitter internal pressure and to protect controls when not in use.

3.4.1.14 Electrical Connections - Connections to external circuits shall be provided as indicated below:

Reference Designation	Receptacle Type (or an approved equivalent)	Function
J-1	Cannon Electric Co. #MC-14H-10-6PN	Power & Test
J-2	General RF Fittings Inc. #182A (Female, Coaxial TNC)	Receiver Connection
J-3	Coaxial TNC, Female (Part of Magnetron)	Transmitter Connection
J-4	Cannon Electric Co. #MC-14H-10-6PN	Connect to Control for Code Selection

3.4.1.15 Test Points

3.4.1.15.1 External Test Point - A pulse signal equivalent to the decoder input signal and indicative of the transmitter reply shall be brought out to external connector for test use while the transponder is operating. The amplitude of this signal shall be at least one tenth of a volt peak into a 470 ohm load.

3.4.1.15.2 Internal Test Points - Sufficient test points shall be provided inside the transponder set to allow rapid isolation of malfunctions to a particular section of the transponder.

3.4.1.16 Pressurization - The receiver-transmitter shall be capable of maintaining an internal pressure of 30 psi above the external pressure. The following provisions shall be met:

(a) The case shall be of a type that will permit ready access for alignment, testing, and repair. The receiver-transmitter shall be fully operable with the case opened, and the tuning adjustments of the receiver-transmitter shall not be affected by opening or closing the case.

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(b) Means shall be provided for pressurizing the receiver-transmitter to 30 psig and measuring this pressure to determine the effectiveness of the seal. An air valve (not part of the radar beacon) shall be used for this purpose.

(c) The seal shall be effective after repeated openings and resealings under field conditions.

*3.4.1.17 Elapsed Time Indicator - An elapsed time indicator shall be provided for the configurations shown in Table I. The elapsed time indicator shall be attached to the R-T unit and shall measure the total time that 28 vdc has been applied to the R-T unit. The indicator shall have a 1000-hour range.

3.4.2 Control C-4419/APN-154(V) - The Control shall meet the applicable requirements of Specification MIL-C-6781 for Type 1 Control Panels.

3.4.2.1 Function - This unit shall provide controls for turning on the equipment and for selecting the mode of decoder operation.

3.4.2.2 Contents - This unit shall contain a toggle switch for turning equipment off and on, a rotary selector switch for selecting the mode of decoder operation and may contain passive circuit components.

3.4.2.3 Form Factor - This unit shall have a maximum allowable size of 5.75 inches wide by 1.5 inches long by 2 inches deep.

3.4.2.4 Weight - The weight of this unit shall not exceed 0.75 pounds.

3.4.2.5 Controls - The following controls shall be provided:

(1) "PWR" (Power) - A three-position switch shall be provided for controlling the power to the equipment. The positions shall be marked "off, "standby" and "operate".

(2) MODE - A rotary selector switch shall be provided for selecting the mode of operation of the equipment. Positions shall be as follows;

(a) Single

(b) Double - 1

Double - 2

Double - 3

Double - 4

Double - 5

3.4.2.6 Electrical Connections - Connections to external circuits shall be provided by means of a standard connector in accordance with Specification MIL-E-5400.

*3.4.3 Duplexer CU-1104/APN-154(V)

3.4.3.1 Function - The Duplexer shall perform the following functions:

(1) Connect the receiver and transmitter antenna terminals of the Receiver-Transmitter in such a manner that the

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radar beacon may operate from a single antenna for both transmitting and receiving. The transmitter and receiver frequency shall be separated by at least 60 mc.

(2) Isolate the rf output of the transmitter from the "long line" effects of the antenna, which might detune the transmitter or excessive amount.

(3) Isolate the receiver input from the transmitter output to prevent damage to the mixer crystal.

*3.4.3.2 Form Factor - The Duplexer shall not exceed 2.16 inches by 1.22 inches in cross section (excluding connectors) and 0.89 inch in thickness.

3.4.3.3 Weight - The weight of the Duplexer shall not exceed 0.5 pound.

3.4.3.4 Electrical Connections - The external connectors shall be coaxial TNC.

*3.4.4 Cable Assembly, Radio Frequency CG-2574/APN-154(V)

3.4.4.1 Function - Two interconnecting cables shall be provided to connect the Receiver-Transmitter to the Duplexer.

3.4.4.2 Form Factor - The cables shall not exceed 6 inches in length and shall be adequate to allow the equipment to meet the requirements of this specification.

3.4.4.3 Weight - The weight of each cable shall not exceed 0.1 pound.

*3.4.5 Control C-4419A/APN-154(V) - Same as C-4419/APN-154(V) except an elapsed time indicator shall be provided. The form factor shall not change.

*3.4.5.1 Weight - The weight of this unit shall not exceed .85 pounds.

*3.4.5.2 Elapsed Time Indicator - The indicator shall be attached to the front panel and shall measure the total time 28 vdc has been applied to the control. The indicator shall have a 1000-hour range.

4. QUALITY ASSURANCE PROVISIONS

4.1 General - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(1) Preproduction Tests

(2) Production Sample Tests

(3) Acceptance Tests

(4) Life Tests

4.2 Preproduction Tests - Preproduction tests shall be made on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall consist of the following:

(1) Contractor's Demonstration Tests

(2) Service Approval Tests

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4.2.1 Contractor's Demonstration Tests - Contractor's demonstration tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.6. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory.

4.2.1.1 Demonstration Test Data - The contractor shall submit all data collected in conducting these tests to the procuring agency for review and approval. On production contracts this data shall include a list of all electrical and electronic parts giving their specified voltage, current and temperature rating and the applied circuit voltage, current and ambient surface temperatures. The ambient and surface temperature shall be obtained under the extreme high temperature operating condition. Such data on parts may be included as part of the preliminary parts list.

4.2.2 Service Approval Tests - At the completion of the contractor's demonstration tests and when requested by the procuring activity, the equipment shall be delivered to a specified government laboratory for additional testing. This additional testing may consist of duplicating tests previously conducted and such other tests as are deemed necessary to determine compliance with all applicable design and performance requirements.

4.2.2.1 Accessory Material - In addition to the complete equipment submitted for service approval tests, the contractor shall also submit the accessory material and design test data specified in Specification MIL-E-5400. This information shall indicate the physical and electrical characteristics of the equipment and establish the equipment's compliance with applicable requirements.

4.2.3 Scope of Tests - Preproduction tests shall include all tests deemed necessary to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with Specification MIL-T-5422 and interference tests in accordance with Specification MIL-I-6181.

4.2.4 Preproduction Approval - Approval of the preproduction model shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction model. Prefabrication of production equipment prior to the approval of the preproduction model is at the contractor's own risk. The approved preproduction model will be returned to the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction model shall not be considered as one of the equipments under the contract; however, it may be reworked by the contractor and submitted for acceptance as a production equipment.

4.2.5 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection and for equipment already accepted by this government it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Production Sample Tests - When requested by the procuring activity, one equipment shall be selected from the first 10 production equipments submitted for acceptance and sent to a designated government laboratory for tests. The equipment shall be selected by the government inspector after the equipment has successfully passed all individual tests.

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4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction, workmanship, performance and quality and that it meets all applicable requirements.

4.3.2 Production Sample Approval - Approval of the production sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the production sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.3 Reconditioning of Production Test Sample - On completion of the production sample test the equipment shall be reworked by the contractor by replacing all wear or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Sampling Tests

(3) Reliability Assurance Tests

(4) Special Tests

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Manufacturing Reliability Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characterisites and record adequate data to assure satisfactory equipment operation.

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*4.4.1.3 Manufacturers Run-in Test - Each equipment shall be operated under the conditions specified herein for a period of 6 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

Temperature	Ambient room
Humidity	Ambient room
Vibration	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ±3g's

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to that plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions the vertical direction shall be used. During the 6-hour period of operation following the 10-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

4.4.2 Sampling Tests - Sampling tests shall be conducted on equipments which have successfully passed the individual tests. Equipments shall be selected for sampling tests by the Government Inspector in accordance with the following: (When reliability assurance tests of 4.4.3 are conducted, sampling tests shall be omitted.)

Quantity of Equipments Offered for Acceptance	Quantity to be Selected For Sampling Test
First 10	1
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

4.4.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

*NOTE: This test shall not be required if a reliability test is selected that requires a burn-in test for all equipments.

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(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, production and special tests.

(3) Manufacturing reliability test specified in para. 4.4.1.3 except that the test duration shall be 120 hours with no restriction on the number of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future equipment.

4.4.3 Reliability Assurance Tests - Reliability Assurance Tests shall be conducted using MIL-STD-781. Tests as required by both the Qualification Phase and the Sampling Phase shall be conducted.

4.4.3.1 Qualification Phase - Prior to the acceptance of equipments under the contract or order, a minimum of three (3) equipments shall be tested as outlined in MIL-STD-781, under the section entitled "Qualification Phase of Production Reliability Tests". The maximum number of equipments to be used shall be those listed in Table 5 of MIL-STD-781. For the Qualification Phase, Test Level E shall be used. The Accept-Reject Criteria for Test Plan II shall be used.

4.4.3.2 Reliability Sampling Phase Tests - The Reliability Sampling Phase test shall be accomplished on 100 percent of all production equipment **not subjected to the Reliability Qualification Phase Test. Each equipment produced shall be subjected to a "burn-in" of two complete cycles as defined in 4.4.3.3. This burn-in may be extended at the option of the contractor. Following the burn-in, the equipment shall be tested for eighteen operating hours (four complete cycles of 4.4.3.3).**

To determine whether the MTBF is being met at any time during the contract the operating test hours and the failures thereon (not counting burn-in failures or burn-in operating time) shall be totaled and the results compared with the reject line of Test Plan II of MIL-STD-781. (Extend the line as necessary to accomodate the data.) These totals shall accumulate so that any one time the experience from the beginning of the contract is included. At the conclusion of each month the test results shall be sent to the procuring activity and to the Naval Air Systems Command, Attn: Avionics Division. At any time that the current totals of test hours and test failures plotted on Test Plan II curves show a reject situation, the procurring activity shall be notified. The procurring activity reserves the right to stop the acceptance of equipment at any time that a reject situation exists pending a review of the contractor's efforts to improve the equipment, the equipment parts, the equipment workmanship, etc., so that the entire compilation will show other than a reject decision.

4.4.3.3 Duty Cycle - The duty cycle to be used shall be in accordance with Test Level E of MIL-STD-781, and shall consist of a 4-1/2 hour heating period (equipment on) followed by a 1 - 1/2 hour cooling period (equipment off).

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4.4.3.4 Test Details - The test details such as the performance characteristics to be measured, special failure criteria, preventive maintenance to be allowed during the test, etc., shall be part of the test procedures to be submitted and approved by the procuring activity prior to the beginning of the Qualification Test Phase of the Reliability Assurance Tests.

4.4.4 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling tests.

4.4.4.1 Special Tests Schedule - Selection of equipments for special tests shall be made as follows; except that when a Production Sample is requested and delivered to a government laboratory for test (see para. 4.3) the requirement for a special test on an early production equipment may be omitted.

(1) On an early production equipment.

(2) On an early equipment after an engineering change.

(3) Whenever failure reports or other information indicate additional equipments should be tested. (This will be determined by the procuring activity.)

4.4.4.2 Scope of Tests - Special tests shall consist of such tests as are approved by the procuring activity. Test procedures previously approved for the preproduction and sampling tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests. As a minimum, the following tests shall be made, except that for those equipments selected as a result of design changes and failure reports, only those tests required to check the characteristics in question need be conducted.

(1) Temperature - Altitude Tests

(2) Vibration Test (60 minutes cycling in each of 3 planes)

(3) Humidity Tests

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4.4.5 Equipment Failure - Should a failure occur during either the sampling or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Life Test - The contractor shall furnish all samples and shall be responsible for accomplishing the life test. The test shall be of 300-hours duration and shall be conducted on equipments that have passed the individual test. The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected by the government inspector in accordance with the following; except that when a production sample equipment is requested and submitted to the procuring activity for test then the first equipment specified below need not be selected or subjected to this test. (Equipments which have successfully passed the sampling tests or special tests may be selected for life tests.)

Quantity of Equipments Offered for Acceptance	Quantity to be Selected For Life Test
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

4.5.1 Test Conditions - The life test shall be conducted under the following simulated service conditions.

Temperature	Normal room
Altitude	Normal ground (0 - 5000 ft.)
Humidity	Room ambient
A.C. Voltage	115 ±5 volts (at lowest applicable frequency)
D.C. Voltage	27.0 ±2.0 volts

4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

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4.5.3 Performance Check - At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.5.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following:

(1) Part type number.

(2) The circuit reference symbol number.

(3) The part function.

(4) Name of the manufacturer.

(5) Nature of the failure.

(6) The number of hours which the part operated prior to failure.

4.5.4.1 Failure Report - In the event of a failure, the government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.5.5 Reconditioning of Life Test Samples - An equipment which has been subjected to the life test shall be reconditioned as follows:

(1) On completion of the life test, the equipment shall be reworked by the contractor by replacing all "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring activity.

(2) After reworking, the contractor may resubmit the equipment for acceptance.

4.6 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test preparation of test procedures. When test procedures are available from previous contracts such procedures may be used when approved by the procuring activity.

4.7 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.8 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the government inspector.

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5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-E-17555.

6. NOTES

6.1 Use - The radar beacon shall be for use in naval aircraft in conjunction with radar direction finding and ranging to increase the range capability of the system.

6.2 Test Values - Normal and limiting values of performance data shall be determined at input voltages of 28 volts ± 1.0 VDC. These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standard of performance.

6.3 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply.

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced herein. All referenced specifications shall apply to the extent specified.

6.4 The parentheses (*), when used in the type designation, shall be replaced by either a number of letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.2.9. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

6.5 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.6 Definitions - The following definitions shall apply:

6.6.1 Interrogation - An interrogation is one or two radio frequency pulses transmitted by an interrogator or radar set.

6.6.2 Reply - A reply is a radio frequency pulse transmitted by a beacon as a result of an interrogation.

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6.6.3 Pulse Width - Pulse width is that interval of time between points at which amplitude of the pulse is 50% of the pulse amplitude.

6.6.4 Pulse Amplitude - Pulse amplitude is the maximum voltage amplitude of the pulse or r-f pulse envelope, excluding spike.

6.6.5 Pulse Rise Time - Pulse rise time is that interval of time required for the leading edge of thepulse to rise from 10 to 90% of the pulse amplitude.

6.6.6 Pulse Fall Time - Pulse fall time is that interval of time required for the trailing edge of the pulse to decay from 90 to 10% of the pulse amplitude.

6.6.7 Pulse Spacing - Pulse spacing is that interval of time between 50% pulse amplitude points on the leading edges of the pulses involved.

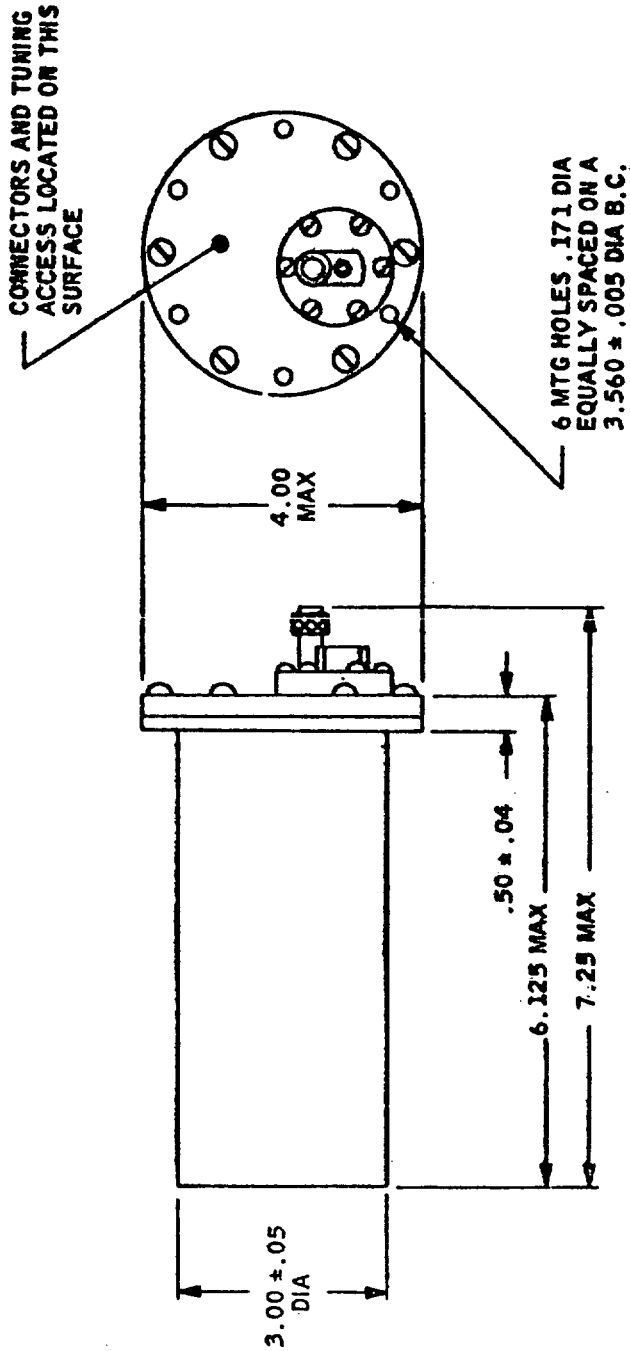
6.6.8 Threshold Signal Level - The minimum signal level, measured at the radar beacon antenna terminal, that causes the radar beacon to reply to at least 99% of the interrogations per second.

6.7 As a general rule non-repairable subassemblies should be encapsulated or hermetically sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts tolerances and ratings should be so selected that the life of the subassembly is greater than that of similar repairable one. With few exceptions (such as high voltage power supplies), the non-repairable subassembly should evidence a Mean-time-to-failure greater than 5000 hours, and for many applications this figure must be nearer 50,000 hours.

*6.8 The contract or purchase order is to specify the configuration of the Receiver-Transmitter unit, the Control, the Duplexer, and the Cable Assembly.

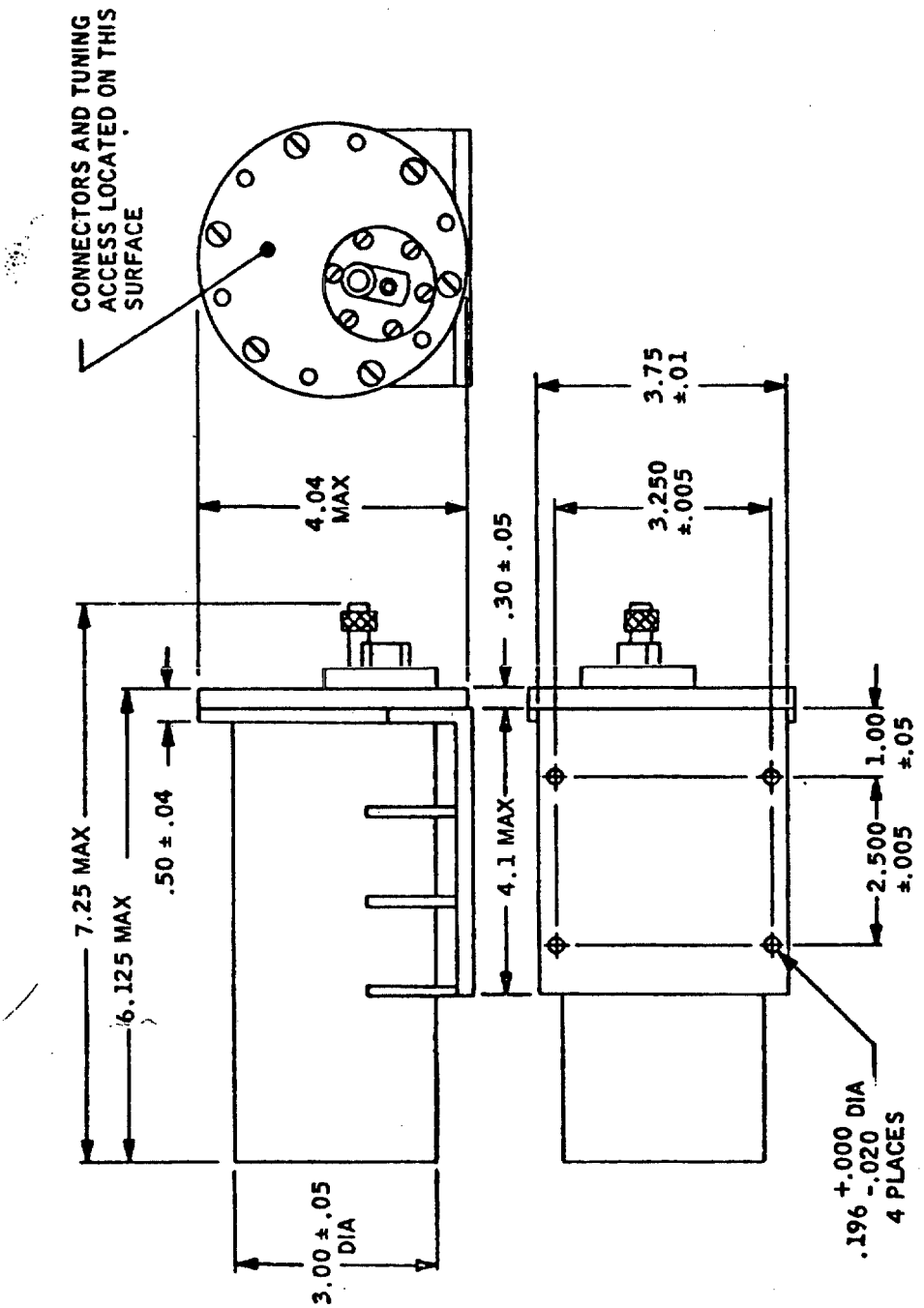
NOTICE - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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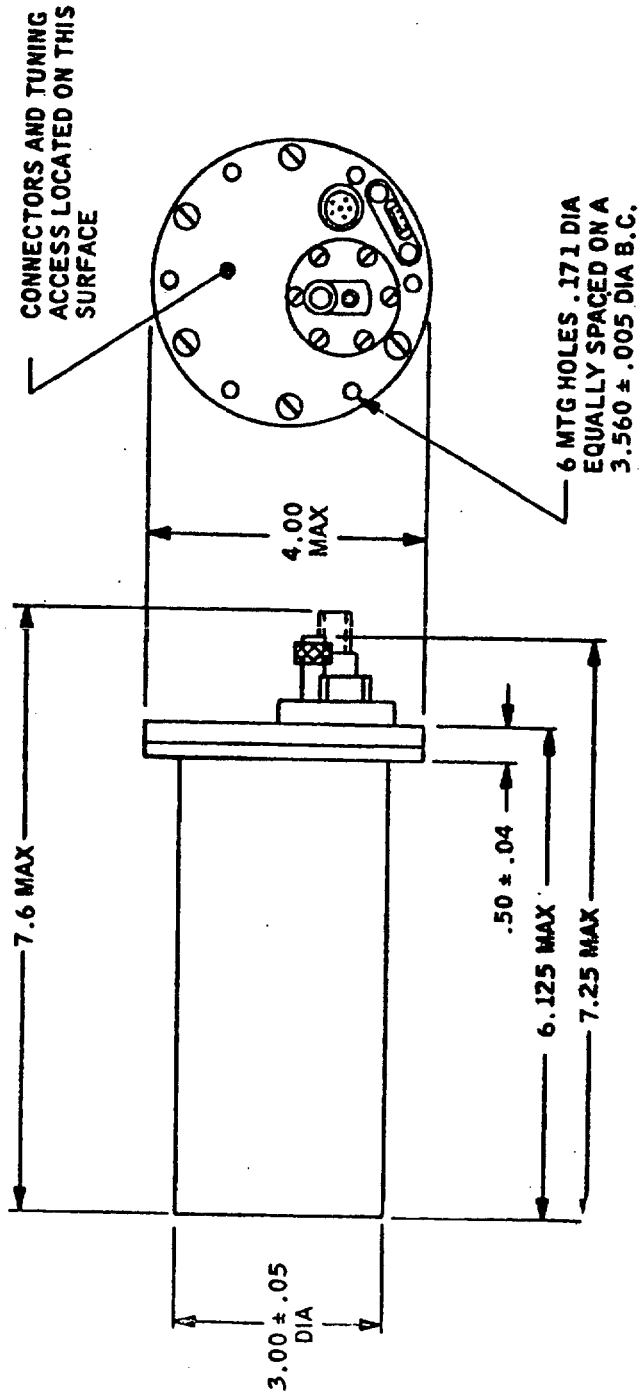
RECEIVER - TRANSMITTER RT-681/APN-154(V)
DESIGN CONTROL DRAWING
FIGURE 1

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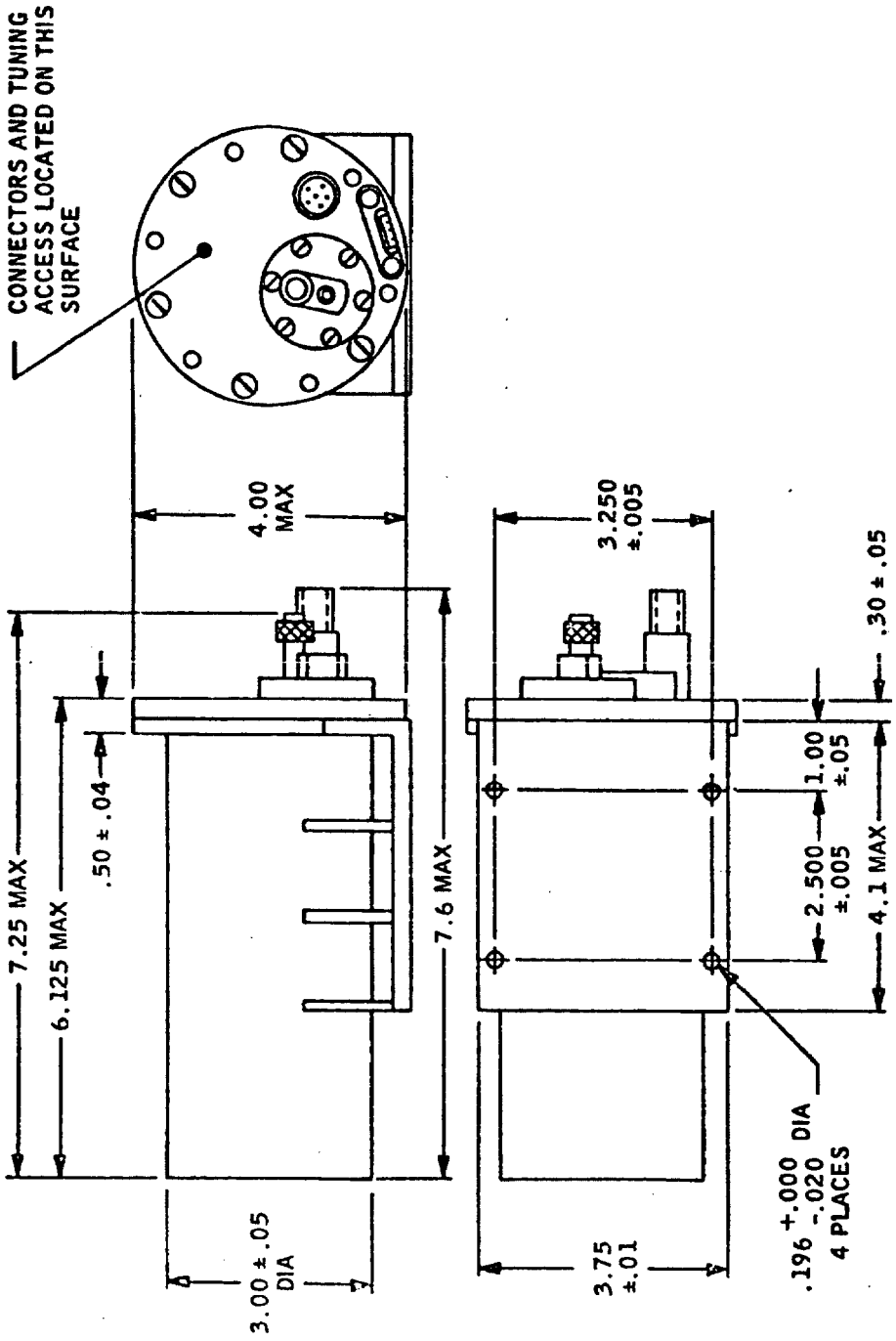
RECEIVER-TRANSMITTER RT-762/APN-154(V)
SIDE MOUNT
DESIGN CONTROL DRAWING
FIGURE 2

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RECEIVER-TRANSMITTER RT-681A/APN-154(V)
FLANGE MOUNT WITH ELAPSED TIME METER
DESIGN CONTROL DRAWING
FIGURE 3

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RECEIVER-TRANSMITTER RT-762A/APN-154(V)
SIDE MOUNT WITH ELAPSED TIME METER
DESIGN CONTROL DRAWING
FIGURE 4

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